

LIGS: Developing an LLM-infused Game System for Emergent Narrative

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Abstract

Games with emergent narratives enable users to craft their own stories through simulation-based mechanics, but have shown various limitations due to the deterministic nature of traditional algorithms. Generative AI, including LLMs (large language models), is proposed as a solution by automating tasks and dynamically generating content tailored to evolving gameplay contexts. While prior studies have explored the applications of LLMs, there is still limited understanding of the challenges and user experiences that emerge when integrating these models into systems. This paper introduces LIGS, an LLM-Infused Game System designed for emergent narratives, and presents a prototype game used to observe actual gameplay experiences and progression. Our findings indicate that participants find the freedom of action and the resulting narrative progression engaging. However, LLM can cause various misunderstandings, posing potential risks to the overall experience of the game. Based on these findings, we propose design considerations to address these issues.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Generative AI, Large Language Models, LLMs, Game, Digital Game, Emergent, Emergent Interaction, Emergent Narrative, User Study

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1 INTRODUCTION

Generative AI has recently gained attention as a technological foundation for enabling emergent narratives [19]. In particular, LLMs have demonstrated its human-level abilities to create diverse content [5, 24] using their foundational knowledge [17] and advanced linguistic communication skills [10], particularly in areas of logic

and creativity [9, 26, 31]. LLMs have also been used in the game industry, such as generating game quests [1, 3] and facilitating NPC interactions [6, 7], and showed their potential to assist or even replace elements traditionally planned and created by humans. Furthermore, research has begun exploring the use of LLMs to build emergent narrative environments [15, 16, 25], such as social simulations [15], further expanding their applicability in the game domain.

Although LLMs have shown potential to generate emergent game environments, their technical implementation and impact on gameplay experiences have much room for exploration and investigation. Previous studies have proposed to leverage AI to enable realistic and open-ended game world systems, often focusing on conceptual data and logical frameworks [1, 3], with the assumption that AI operates under ideal conditions. However, the performance of LLMs, which is affected by bias, prompts, hyperparameters, and other factors, often leads to unexpected issues and results [28]. For example, biases have been observed in the generated endings, favoring overly positive resolutions [22, 23]. This divergent and difficult-to-control nature of AI can lead to unpredictable gameplay experiences [30]. Therefore, to effectively apply AI, systems must be designed to consider its characteristics and performance, while mitigating risks and inconsistencies through a deep understanding of LLM functionality and user interactions.

This paper explores the potential of emergent narrative games, focusing on the possibility of interaction and storytelling via LLMs, and tries to identify associated design and technical challenges. We present LIGS, an LLM-Integrated Game System, where players can freely input text to control the Player Character (PC), while LLMs dynamically operate Non-Player Characters (NPCs) and update in-game objects as the narrative unfolds. We conducted a user study of semi-structured interviews with 20 participants, followed by a qualitative analysis of the collected data. By examining how an LLM gradually evolves stories and the game world in response to user interactions, our study examines how LLMs influence gameplay dynamics and user experiences by analyzing their role in generating emergent narratives and adaptive game worlds, highlighting both the opportunities and challenges they present.

2 RELATED WORKS

Emergent narratives in games provide the advantage of responding flexibly to the actions of players, facilitating unique and dynamic storytelling experiences [2, 12, 21]. A prominent example is *The Sims*, which simplifies real-life social dynamics to allow players to create narratives as alternate selves in a virtual world [14]. Despite the obvious benefit, emergent game systems face several challenges

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[19], including the need for vast modular content, often constrained by the repetitive nature of procedural generation, and the absence of algorithms capable of simulating emotions or internal states, which limits the reuse of manually created assets. Further challenges are pointed out [19] in the study, noting the interpretation of the narrative is left to the player due to its reliance on dynamically generated events. This makes it difficult to perceive events as part of a coherent narrative.

Generative AI technologies, particularly LLMs, are poised to address the constraints of traditional emergent narratives [19]. Beyond predicting the next word in a sequence, LLMs exhibit logical reasoning [8], inference capabilities, and knowledge derived from extensive datasets [10]. Advancements in models such as GPT-4, Llama, and Claude have significantly mitigated former limitations such as limited context windows and hallucinations, paving the way for more structured and diverse narrative generation [13, 27]. With capabilities such as advanced text generation, persona-based role-playing [20], and contextual understanding [10, 11], LLMs enable the creation of human-level narratives. These capabilities offer solutions to existing challenges and are expected to play a key role in enhancing player experiences and broadening narrative possibilities in games. Numerous attempts have been made to create a realistically free game world by leveraging artificial intelligence. For example, StoryWorld [18] procedurally constructs quests, terrains, and character histories to provide a novel gameplay environment each time. CONAN [4] generates quests by considering the intentions of the NPCs and the state of the game world. Ashby et al. [3] suggested a framework for quest and dialogue generation for a grounded and personalized experience. However, these studies focus on the generation of discrete outcomes, with limited attention to the dynamic nature of games as media that continuously progress and evolve through player interaction. Consequently, identifying unforeseen situations remains challenging, highlighting the need for research that integrates real-world development contexts.

Building on these efforts, further research has explored LLMs to generate emergent responses and narratives in interactive game environments. Recent research has explored the use of LLMs to generate emergent responses. Yin et al. [29] designed a game where NPCs make false statements with LLMs and examined how these false statements influence the gaming experience. Their findings provided implications for game design and insights into players' perceptions of LLM usage. Peng et al. [16] applied LLMs to a text adventure game, allowing for freeform user interactions and facilitate emergent narrative possibilities. However, these studies were constrained by a fixed interaction framework, limiting narratives to the game's predefined structure. Further, LLMs have been explored for generating emergent narratives. Park et al. [15] developed LLM-driven agents with cognitive capabilities (memory, planning, reflection) and action, and demonstrated that these agents demonstrated human-like behaviors in a virtual village simulation. Wang et al. [25] suggests LLM-driven narrative framework that co-created by the author, the simulated characters, and the player. However, these studies have not examined actual user experience with these framework as a game. While Yong & Mitchell [30] conducted a user study on ai dungeon and showed that excessive freedom or unpredictable responses can reduce player engagement, this is more closely related to basic usage of LLMs and interactive narrative

experiences, creating a gap compared to traditional games, this highlights the need for a deeper understanding of LLMs in game design to address their limitations.

Therefore, research on how emergent responses in game leads dynamic changes in narratives and the game world, as well as the user experience, is yet fully discussed. In this paper, we generate dialogue, action, and resulting narrative progression through LLMs. Through this process, we explore user experiences and the technical challenges of LLM integration through comprehensive user studies—such as interpreting hallucinations as part of the feedback on game progress or addressing inability to prioritize choices due to an ambiguously defined game world. By examining those overlooked aspects, we seek to uncover key insights necessary for unlocking the potential of generative AI in game design, advancing the possibilities of immersive and adaptive gameplay.

3 LIGS: LLM-Integrated Game System

We propose LIGS (LLM-Integrated Game System), a game system that leverages LLM as its core engine to facilitate user interactions and manage emergent narrative. In LIGS, players can freely describe their player character's actions in natural language, while the LLM determines actions of non-player characters (NPCs) and responding stories. The LLM then updates the game's world data based on these interactions. By developing LIGS, we aim to identify its opportunities and challenges and use it as a research tool to gain insights into its practical implementation and user experience.

Game World Data Structure. As shown in Figure 1, game world is composed of four classes, each designed to fulfill specific roles within the game and inheriting from the base structure of object: object represents all entities; space serves as the highest-level parent, defining areas (e.g., rooms) that can contain other objects; character describes entities capable of autonomous perception, reasoning, and action; and player represents agents controlled by the user. Every instance has textual properties (type, name, description), which define and describe the object (e.g., a "furniture" named "wardrobe" with "the door is closed, blood is seeping out"). Every instance also has relational properties (relation, child), which describe and connect objects through a hierarchical structure (e.g., "wardrobe (child) is inside the library (parent)"), with at the top of the hierarchy is a space. This structure enables the game world to be clearly represented and organized, allowing both game designers and LLMs to define, understand, and update game world.

NPC Simulation. LIGS provides NPCs with spatial and event information, allowing them to determine actions through their mental processes. These actions are relayed back to the system, ensuring autonomy while maintaining narrative consistency. This study does not focus on simulating characters, instead simplifying the processes to essential elements for plausible progression to reduce cost and generation time during user testing. Characters have three variables for internal process: personality, self-reminders, and sequential memory. Personality is represented by simple keywords, while self-reminders are static paragraphs describing the character's perception of their situation. Sequential memory stores generated thoughts for use in future internal processes. First, the

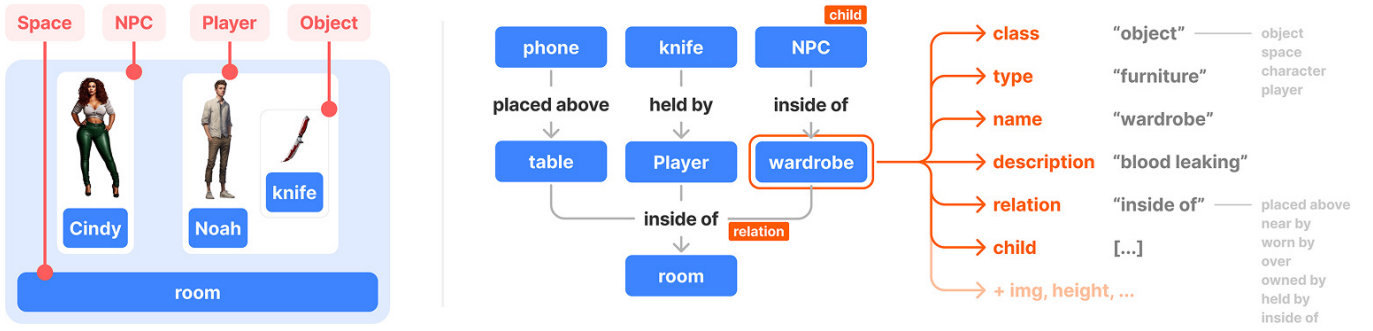


Figure 1: Types (left), Data structure and the hierarchical structure of objects (right) in LIGS

LLM generates the character’s perception of their environment and recent events (recognition), then makes predictions and determines actions (prediction and action). These recognitions, predictions, and actions are stored as sequential memory for future reference.

Event Generation. Using LLMs, the system aggregates characters’ actions for the current turn and generates the next event in a narrative format, resolving conflicts and describing interactions. Events are categorized into four types: Conversation, Spatial Movement, Parent Movement, and Action. This systematic approach ensures a detailed, clear, and coherent narrative while handling multi-character interactions. When a conversation occurs between characters, the system generates a script-like output, including names, simple emoji codes, basic actions, and dialogue. For object movement, it describes what has moved and where it has moved to. In the case of parent movement, it explains what has gained a new parent and their new relationship. The distinction between these two types of movement is entirely delegated to the language model. Finally, the system describes the character’s actions and their resulting outcomes. The model updates the game world by identifying changes in game objects based on generated events. It lists updates needed for object information or locations, summarizes descriptions or movements (spatial, parent) accordingly. The system utilizes this structured data to ensure consistency and accuracy when updating the game world.

Game Process. The game progression involves three steps: players review textual information about events, select objects and actions and gather information, then submit their choice to the system and await the generated results. To enhance clarity, the game world data is visualized as a hierarchical list. Each objects show their image and name. Spatial objects appear as full-screen backgrounds, with child objects displayed on their name box. Relationships are arranged intuitively: “above” relationships are stacked vertically, while others are aligned horizontally relative to the parent. Although object images were not a primary focus, they were included for readability and quick identification for only pre-defined objects. The input interface combines both action choices and a text field, allowing for flexible user interactions while examining how each interface impacts the user experience and game progression. These elements are commonly used in text-based games and interactive narratives powered by LLMs. When an object is selected, the system generates three action choices based on the object and the

current context. Additionally, players can freely enter actions for their character, directly in the text field regardless of the selected object.

In The Cabin: A playable prototype game of LIGS. Based on the system, we developed a playable mystery genre game prototype - In The Cabin - to explore the possibility and limitation of the LIGS. The mystery genre was chosen for its realism, supporting diverse interactions in confined spaces and aligning with the study’s resources. Action-heavy genres were excluded due to challenges with real-time LLM interactions, and fantasy or horror were avoided due to difficulties in defining clear rules when generation. The story involves four characters: Noah (player), John, Amy, and Cindy. John and Amy are a couple, while Cindy secretly loves John. Cindy kidnaps Amy and locks her in a cabin, where the game begins with Noah and John searching for her. The cabin includes two areas, the living room and reading room, with various objects to enable interactions and scenarios. To enhance immersion, the player is not initially informed of the characters’ relationships, receiving necessary information only upon request.

4 USER STUDY

A user study was conducted to examine the experiences and perceptions of both gamers and experts regarding the prototype, “In The Cabin.” The methodology involved gameplay sessions followed by interviews to gather qualitative data. Researchers observed how the LLM-driven system managed emergent narratives, providing insights into its potential and challenges. The study recruited 8 gamers and 12 game development experts, after excluding one expert who were unable to participate fully in the experiment. Experts, including developers, directors, and scenario writers, were chosen for their contributions to key elements of the study’s focus — such as system architecture, data structures, narrative creation, and world building. Gamers were recruited through university forums, while experts were recruited via university and game creator forums, and snowball sampling. Gamers focused on narrative games, with those playing primarily non-narrative genres excluded.

Semi-structured interviews were conducted to gather comprehensive results. Gamer interviews lasted one hour in two sessions, while experts had additional one session totaling two hours. All interviews were one-on-one, except for three experts who requested a joint session. Participants were briefed on the study details prior

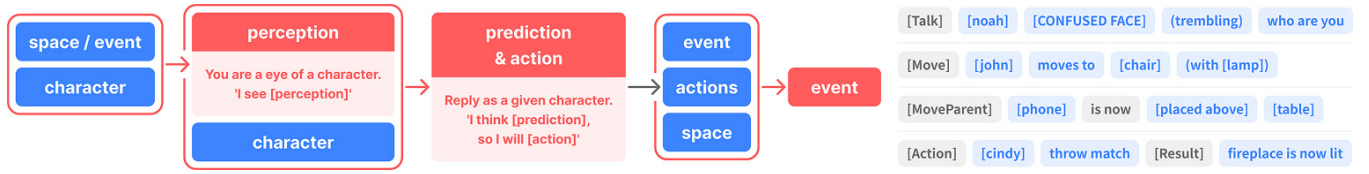


Figure 2: NPC inner process (left) and Event generation process (right)

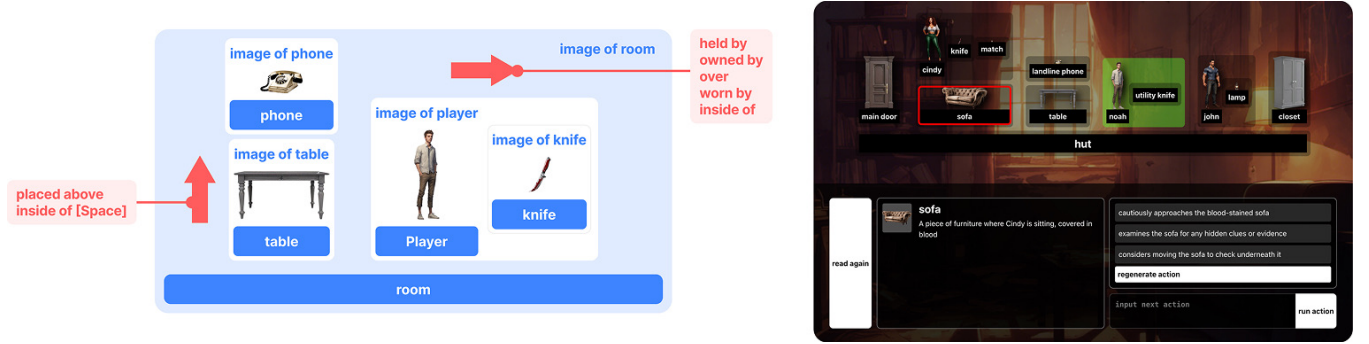


Figure 3: Visual representation of positional relationship (left) and In game screenshot (right)

to its commencement, and their consent was obtained for audio and video recording. The study was approved by the Institutional Review Board (IRB) of the researcher’s affiliated institution. *Introduction (15 minutes)*: The interviews began with an introduction to the study and an icebreaker about participants’ gaming experiences, including their favorite games and preferences, and their experiences with AI tools. Experts also shared insights from their roles in game development. *Game Prototype Experience (55 minutes)*: Participants freely played game and shared feedback on satisfaction, immersion, narrative personalization, and interactions with in-game elements. They also discussed challenges, replayability, and the ease of input composition. *Expert Session (40 minutes)*: Experts reviewed game data, systems, and UI materials, offering feedback on guidance, functionality, and narrative planning. The session concluded with discussions on improving the prototype and the potential of AI-integrated games in the industry.

A qualitative analysis was conducted using all experimental outputs, with interviews transcribed via CLOVA Note and verified by the researcher. Data were analyzed through thematic analysis, involving iterative review, coding, and organization by the primary researcher, with findings reviewed and refined by a secondary researcher. The data was organized using the criteria below to focus on user perceptions and the broader implications of LLM applications in gaming. *Focus on Participant Perception*: While previous studies focus on the functioning of emergent narratives, this study examines how users perceive system-generated narratives, emphasizing emotional engagement, coherence, and agency to enhance the understanding of emergent narrative experiences. *Exploration of LLM-Driven Experiences*: This study examines the unique experiences and challenges of integrating LLMs into game systems, including emergent narrative structures and unexpected user interactions. By comparing user-reported experiences with intended

game data, the research seeks to understand how these phenomena influence user perception and game progression.

5 FINDINGS

5.1 Freedom Enhances Narrative Space, but Challenges Decision-Making

Participants positively evaluated the system’s freedom of actions. E10 explained, “The best part is that there are no restrictions. And it feels like, ‘I just tried this, and it actually works.’” Compared to traditional games, where objects are restricted to predefined uses, this game was highly appreciated for the flexibility it provided. G3 stated, “What I liked the most was that I could decide how to use each object.” The game’s ability to provide realistic outcomes further enhanced this sense of freedom. G3 commented, “I felt like the actions I took were reflected in a very realistic way, which seemed to shape the game’s story well.” Participants also described the narrative progression as highly unrestrained. G2 commented, “It felt like a very free-flowing story; I never felt like I was just following a predetermined script.”

However, this freedom also caused confusion in decision-making, such as participants struggled to prioritize actions in the open environment. E8 remarked, “There’s so much to interact with, so I was confused about what to prioritize to make the right choice.” The unclear objectives or absence of definitive feedback made decision-making more challenging. E5 explained, “I couldn’t tell where I was in the progression—like, am I in the early stages or the later stages?” They also felt there was insufficient information to understand the situations. G5 commented detailed descriptions on characters’ expressions or actions would have made it easier to deduce situations, and E10 expressed hesitation in taking action, stating, “I don’t know if John is in front of me, if I can reach the knife behind him, or if the knife is closer to me.”

5.2 Emergent Content Drives Flexibility, but Risks Narrative Disruption

The system dynamically generated new attributes and objects. For example, a wardrobe initially defined as “closed” could become openable or locked depending on the narrative. Similarly, an engagement ring observed by the player gained an engraving during the process. Entirely new objects, such as books or paperweights, were also created to fit the context, as in a reading room. G3 remarked, “Seeing things like books or paperweights being generated for some reason made me realize how a highly flexible game could be.” Additionally, exploration revealed new items, such as a key taped under a desk in E8’s gameplay, demonstrating how spatial exploration led to emergent clues and objects.

Players perceived the objects and their attributes, generated during the story creation process, as intentional gameplay design. For example, E5 noted, “I was just thinking of a table, but then it turned out I could open a drawer of table, and there was a ring inside. It felt like these developments were clues leading to certain outcomes in the game, and I found that enjoyable.” Similarly, the success or failure of actions and character reactions were interpreted as guiding the narrative. G8 remarked, “I often felt like the story was already decided, and I just happened to take actions or give responses that fit the narrative they wanted to progress.” G7 pointed out the ambiguity in certain situations, saying, “I was confused about whether I was trying to open the wardrobe at a stage where it couldn’t be opened or if I needed to use a different tool to open it—it felt unclear.”

5.3 Allows Autonomous Play, but Results in Fragmented Narratives

Participants expressed a desire for the game to make them the protagonist of the story. E11 stated, “Ultimately, a game like this is a single-player book. So I think it’s a story for me to become the protagonist.” Participants positively evaluated the experience of constructing the narrative autonomously. E5 commented, “The story didn’t branch out from predefined choices to a conclusion. Instead, I kept creating variables, which made it feel free.” Participants set their own goals and became immersed in pursuing them. G5 shared, “In the process of trying to catch Cindy, I set that as my goal, and maybe that’s why it aligned so perfectly with the objective. It helped me stay immersed.”

Participants noted a lack of player role in the game’s narrative. G8 remarked, “It felt like Noah didn’t have a role. Noah seemed more like an observer.” Participants also expressed frustration at the narrative’s lack of responsiveness to their actions. G8 stated, “I threw a lamp at Cindy, but she just ignored me and kept yelling at John. I thought, ‘I threw a lamp, so shouldn’t she also react to me or threaten me?’” In addition to the lack of player agency, participants highlighted issues with the narrative’s depth and cohesion. E7 remarked, “It feels like there’s no clear intention behind it; it just throws things out based on the situation at hand.” Participants also criticized the lack of cohesion in the narrative. E12 shared, “I tried to do something, hoping for a connection... but it just felt fragmented. It made me think, ‘Is it even listening to me?’ and that broke my immersion.”

5.4 Enhances Automation in Design, but Requires Greater Oversight and Regulation

Participants evaluated AI as capable of handling tasks autonomously based on pre-learned information. E8 commented, “It could make things a lot more convenient. It would handle things without needing to set up everything manually.” Participants expected that AI could reduce design complexity by autonomously managing tasks and information that previously required manual definition. E10 remarked, “If AI can work autonomously with clear coherence, like effortless pleasure without responsibility, this would drastically lower the difficulty of design.”

However, participants expressed concerns about the use of AI in game systems. E9 stated, “The biggest concern is that we need to prevent things we can’t predict,” while E10 described it as “loosening control but still controlling it.” As a result, participants questioned whether using generative AI would truly be efficient. E9 remarked, “We’re crafting more sophisticated and complex prompts to build safety measures, but even then, we can’t guarantee that nothing will go wrong.” Additionally, participants raised concerns about ambiguous standards for handling explicit actions. E8 remarked, “We need to be cautious about how realistic the virtual characters are.” Additionally, E9 questioned how these issues would impact game ratings and policies, stating, “Games need to go through content ratings, so I wonder how this will be addressed by both the reviewers and those submitting for review.”

5.5 Dynamic Gameplay Experience, but Creator Certainty Diminishes

Participants highlighted the game’s ability to deliver a heightened sense of immersion and personalized experiences as key differentiators. E12 described it as “an experience where you can think, ‘That really happened back then,’” They also expressed excitement about the potential for AI to extend playtime and create endless, cost-effective experiences, with E7 noting, “I hope it becomes a game I can play all day.”

However, participants expressed concerns that using LLMs might make it harder for developers to convey their intended experiences. E8 explained, “I think of a game as a complete package, but with this approach, it’s challenging to define a clear conclusion.” Additionally, participants noted that the systems might fail to engage players. E9 explained, “There could be small moments of enjoyment, but it’s questionable whether it could provide the deep engagement needed for long-term play.” Further, participants discussed the impact of AI on the gaming industry, addressing both opportunities and concerns. E3 noted, “Once a game is created, reproducing it will become extremely easy.” E4 further warned, “AI will continue to handle production, leaving humans to only consume.” E11 countered “AI’s purpose and potential depend on human use,” emphasizing the need for effective utilization. E12 added, “Creators should focus on designing environments and conditions to maximize diverse user experiences,” highlighting new creative opportunities.

6 DISCUSSION

Traditional games achieve clarity through the careful design of elements like variables, systems, narratives, and user experience.

These components are logically structured to help players understand the system, adapt to challenges, and make informed decisions. In contrast, LLM-powered game worlds rely on linguistic expression, blurring the boundaries between mechanics, variables, and narratives. This allows dynamic interactions that foster creativity and spontaneity, but creates uncertainty, especially for players used to structured gameplay.

The study reveals that the lack of clear definitions in LLM-based systems significantly impacts player experience. Objects and variables, defined solely through language, often lead to incomplete or evolving descriptions due to the model's iterative revisions. Dynamically generated outcomes, based on contextual plausibility, make decision-making opaque and force players to continuously adjust their understanding and expectations. This ambiguity further challenges players, as they struggle to differentiate between intentional design and emergent outcomes.

To address these challenges, game worlds require clear definition of variables, mechanisms, and AI boundaries. LLMs can function within well-designed predefined frameworks, much like TRPG that provide structured yet flexible worlds. Narrative manager can be applied to bridge the gap between emergent storytelling and structured design, ensuring a polished user experience. Though beyond this research's scope, it highlights the need for new definition of developers' roles, policies and standards. As LLM-based games become more prevalent, these shifts will redefine how games are developed, experienced, and understood.

7 CONCLUSION

This paper introduces LIGS, an LLM-integrated game system, to explore the characteristics, potential, and limitations of LLMs in emergent narrative games. A prototype was developed, and user experiences, gameplay data, and expert feedback were collected through semi-structured interviews. This process revealed both the novel opportunities and challenges of integrating LLMs into narrative-driven games. Our findings suggest that integrating LLMs into game systems enables freeform experiences distinct from traditional games but also introduces potential misunderstandings. To ensure effective implementation, it is crucial to clearly define the roles and limitations of LLMs rather than viewing them as all-encompassing solutions. User perception of the interface and narrative must also be prioritized. Additionally, societal and policy directions regarding LLM applications remain unclear, requiring careful deliberation in the future.

The limitations of this study underscore important areas for future exploration. LLM performance led to challenges such as hallucinations, incomplete context reflection, and unexpected behaviors, while limitations like context window size and generation time also significantly influenced user experience and outcomes. Additionally, The constrained mystery/thriller scenario limited generalizability, while ethical issues, overlooked due to intentional norm-breaking, require further study. Future research should explore broader scenarios, genre impacts, and long-term gameplay factors. Future work will focus on proposing detailed design guidelines and practical recommendations to address identified issues. Future studies will incorporate expert feedback to improve system usability, providing practical insights such as taxonomies of user

behavior and understanding or real-world applications to refine functionality.

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A Participants

Table 1: Demographics of Expert

Participant	Gender	Expertise
E1	Male	Developer
E2	Male	Developer
E3	Male	Developer
E4	Male	Developer
E5	Male	Developer
E6	Male	Director, Developer
E7	Male	Director, Developer
E8	Female	Director
E9	Male	Director
E10	Male	Director
E11	Female	Director, Scenario Writer
E12	Female	Scenario Writer

Table 2: Demographics of Gamer

Participant	Gender
G1	Male
G2	Male
G3	Female
G4	Female
G5	Male
G6	Male
G7	Female
G8	Female

B Game Prototype

For game prototype, there is one version for user research (used for current study), and revised version for deployment we are working on. While the core system remains unchanged, we made two changes for minimizing resource usage: 1) adding a turn limit and an ending generation feature, and 2) merging spaces into one. The game screenshot used in the paper is based on the deployment version we are currently working on, with the only difference from the user research version being small variations in the objects. For deeper look into the research version, refer to the section below (Figure 6, 7).

B.1 Play Procedure



Figure 4: Procedure of the Game: From the left: Read event (left), Explore objects and decide actions (center), Wait for generation (right)

B.2 UI

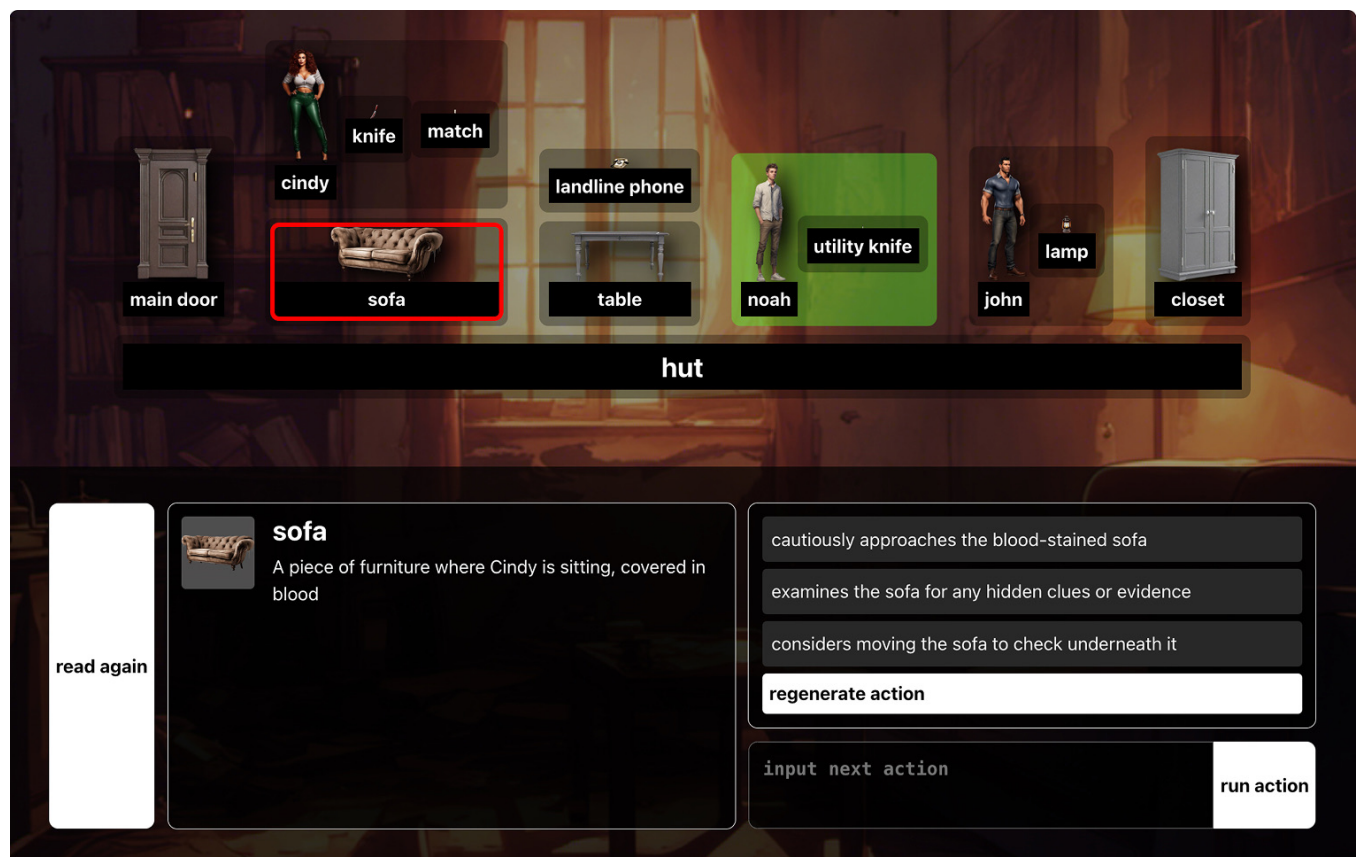


Figure 5: UI Screenshot of the Prototype

B.3 Game World Setting Used for User Research

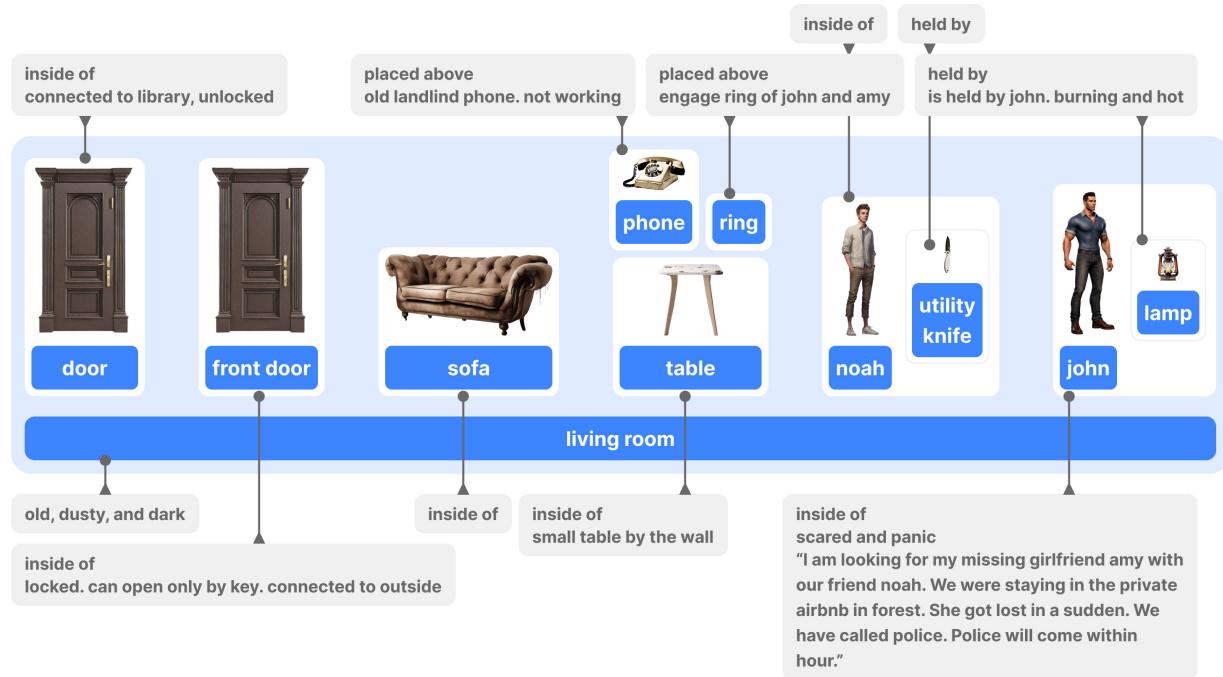


Figure 6: Game World Data of Living Room

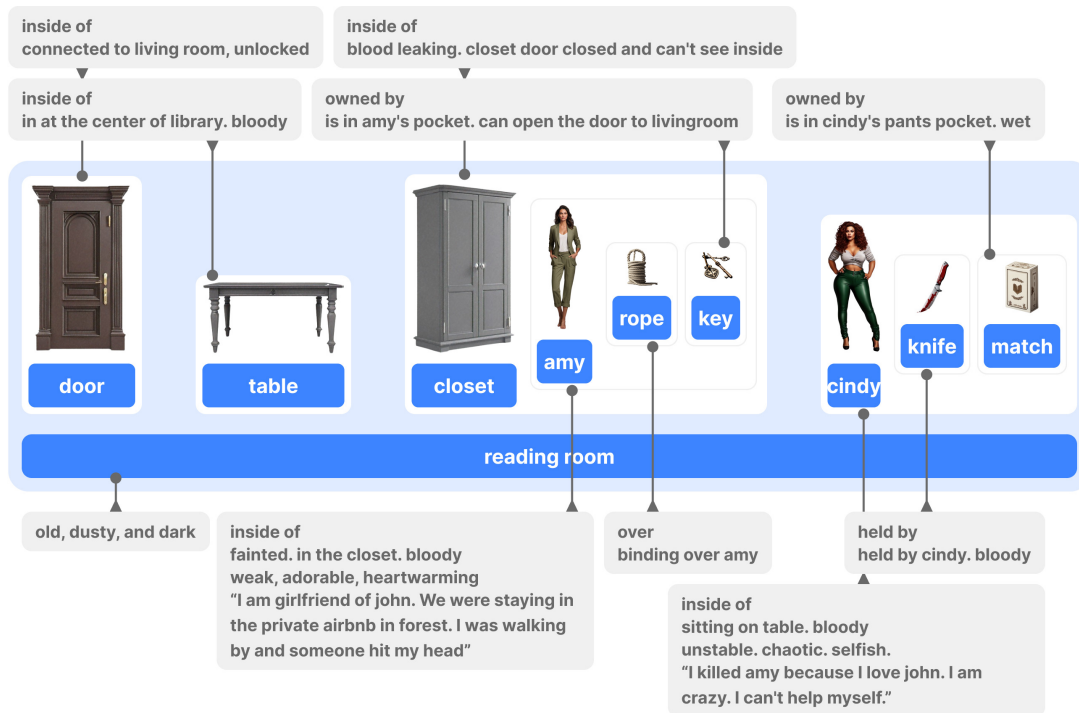


Figure 7: Game World Data of Reading Room

B.4 Example Gameplay

Example gameplay result is based on the version for deployment we are working on. It is added as an example to help understanding the gameplay process. The original gameplay was conducted in Korean and translated into English by a translator, and later reviewed by researcher.

B.4.1 Common Introduction. You are Noah. You, along with your friends John and Amy, have traveled to a cabin in the forest. Late at night, you discover that Amy has gone missing. You searched around the house but couldn't find her. Leaving the panicked John behind, you call the police. The police tell you they will arrive within an hour. Meanwhile, John finds some of Amy's belongings and starts tracking her trail. You follow him as he runs frantically. You and John discover an abandoned cabin nearby. You and John enter the cabin.

B.4.2 Gameplay1. : bad ending scenario

Player Action: Peeking Carefully Under the Table

Cindy charges at John with a bloodied knife. John stumbles backward and barely avoids the blade.
[John] (stepping back) "Cindy, what are you doing?! Stop!"
John walks toward the closet.
Noah carefully peeks under the table. Noah finds nothing special under the table.
[Noah] (with a confused expression) "There's nothing here..."
Amy, struggling to regain consciousness, weakly calls out. A faint sound comes from inside the closet.
[Amy] (whispering) "John... help..."

Player Action: Searching Through the Sofa Cushions

Cindy swings the knife toward John's chest. John uses a lamp to block the knife, producing a loud clang.
[John] (staggering back) "Cindy, stop! What happened to you?!"
John approaches the closet doors.
John frantically tries to open the closet door. The closet door is still locked, and his attempts fail.
[John] (shaking the doorknob) "Amy! Amy, are you in there?"
Noah searches through the sofa cushions. Noah finds nothing special in the cushions.
[Noah] (standing up) "There's nothing here either. What's happening?"
Amy struggles against the ropes binding her. The ropes loosen slightly, but Amy is still bound.
[Amy] (whispering) "Almost there... just a little more..."

Player Action: Knocking Over the Closet with John to Break It

Cindy charges at John with the knife, aimed at his chest. The knife grazes John's arm, tearing his sleeve and leaving a shallow wound.
[John] (grimacing in pain) "Ah! Cindy, please stop!"
John approaches the table.

Noah pushes the closet hard. The closet wobbles but does not topple, and a loud thud echoes through the room.

[Noah] (struggling) "It's too heavy! John, help me!"
Amy kicks the closet door. A dull sound is heard, but the door doesn't budge.

[Amy] (weakly shouting) "Help me! I'm here!"
John swings the lamp at Cindy's head. Cindy ducks and avoids the lamp, stumbling backward.

Player Action: Pushing the Closet Over with John to Crush Cindy

Cindy charges at John with the knife aimed at his heart. John barely dodges, and the knife grazes his shirt.

[John] (staggering back) "Cindy, please! Stop this madness!"

Noah moves toward John.

Noah grabs John's arm and pulls him toward the closet. John breaks free from Cindy's grip.

[Noah] (shouting) "John, help me push the closet!"
Amy repeatedly kicks the closet door. The closet shakes violently, and items spill out.

[Amy] (screaming) "John! Noah! I'm here! Help me!"
Cindy charges at John and Noah. Cindy's momentum is blocked by the tilting closet.

Player Action: Fully Knocking the Closet Over to Crush Cindy

Cindy charges at John, knife raised and aiming for his chest. John barely dodges, but the knife cuts through his shirt and leaves a shallow wound on his side.

[John] (grimacing in pain) "Cindy, stop! This isn't love, it's madness!"

Noah moves toward the closet.

Noah pushes the closet with all his strength. The closet tilts dangerously but doesn't topple, creaking under the strain.

[Noah] (shouting) "John! Help me flip this over!"
Amy fumbles trying to pull a key from her pocket. Amy's fingers grasp the key, but it's difficult to manage in the tight space.

[Amy] (voice muffled) "I'll be there soon! Just a moment!"

John approaches the closet.

John pushes the closet with Noah. With combined strength, the closet tilts further, on the verge of falling.

The rope is inside the closet.

Cindy charges toward John and Noah. Cindy's charge is blocked by the falling closet.

Bad Ending

As John and Noah try to push the closet over, Cindy, using the last of her strength, charges at John. The knife pierces John's chest just as the closet falls, crushing Amy beneath it. The bloodstained closet door opens, revealing Amy's lifeless body. Cindy, grinning madly, collapses on top of John. Noah

stands frozen in horror, staring at the scene. In the distance, the sound of sirens can be heard, but it's already too late. Everything in the cabin has been destroyed, and the dreams and hopes of the four people are shattered. The End.

B.4.3 Gameplay2 : good ending scenario

Player Action: Confessing Hidden Feelings to Cindy

Cindy moved toward the center of the room. She slowly got up from the sofa, gripping the bloodied knife even tighter. As her weight lifted from the cushion, the sofa shifted slightly. Noah moved toward Cindy. [Noah] (nervously approaching Cindy) "Cindy, I... I have something important to tell you." John moved toward the closet. John carefully approached, keeping his eyes fixed on Cindy. The old wooden floor creaked under his footsteps. Inside the closet, Amy struggled to regain consciousness. A faint rustling sound came from within as Amy moved. [Amy] (weakly whispering) "P-please... someone... anyone..."

Player Action: Holding Cindy's Wrist and Confessing Love

Cindy took a threatening step toward Noah, holding the knife firmly. Instinctively, Noah stepped back, fear evident in his eyes. [Cindy] (gritting her teeth) "Stay back, Noah. I won't hesitate to use this." Despite the threat, Noah reached out and grabbed Cindy's wrist. For a brief moment, her grip on the knife loosened in surprise. [Noah] (voice trembling) "Cindy, I... I love you. I always have." John moved toward the closet door. With a lamp in one hand, he reached for the doorknob. As he opened the door, a loud creak echoed through the room. Amy struggled against the ropes binding her. The ropes loosened slightly, but Amy remained tied up. [Amy] (a little louder) "Please... help me..."

Player Action: Kissing Cindy's Hand

Cindy tried to break free from Noah's grip and dash toward the closet. Cindy's sudden movement made her stumble, causing the knife in her hand to waver. She moved toward the closet. [Cindy] (shouting) "No! I won't let you ruin everything!" Noah tightened his grip on her wrist and pulled her back. She staggered, nearly falling. [Noah] (desperately) "Cindy, please stop! This isn't you!" John flung the closet door open, revealing the inside. Light flooded in, illuminating Amy, bound and injured.

[John] (panting) "Oh my God, Amy!"

Amy squinted at the sudden brightness and tried to move toward John. As Amy shifted, pain from her head injury made her wince.

[Amy] (weakly) "John... help... please..."

Player Action: Dancing with a Knife While Singing a Love Serenade

John reached into Amy's pocket to retrieve the key. His fingers brushed against the cold metal. [John] (whispering urgently) "Hold on, Amy. I'll get you out of here." Cindy suddenly mustered her strength, trying to break free from Noah's grasp. Cindy's unexpected struggle made Noah stagger, momentarily loosening his grip. She rushed toward John. [Cindy] (screaming) "No! Don't touch her!" Noah desperately tried to hold onto Cindy. His fingers slipped against her arm, barely managing to keep his grip. [Noah] (shouting) "Cindy, stop! Please!" Amy shifted slightly to make it easier for John to reach the key. As Amy moved, the ropes scraped against her skin, making her let out a pained gasp. [Amy] (gritting her teeth) "Hurry, John... she's coming..."

Player Action: Hugging Cindy While Singing a Love Song

John quickly snatched the key and raised the lamp defensively. Now, the key was in his hand, and the lamp stood between him and Cindy. John moved toward Amy. [John] (speaking quickly) "Got it, Amy. Just hold on a little longer!" Cindy lunged at John with her knife. The blade sliced through the air, barely grazing John's arm. The knife remained in Cindy's hand. [Cindy] (screaming) "You can't take her away from me!" Noah desperately tried to pull Cindy back. His arms wrapped tightly around her waist, slightly restraining her movements. [Noah] (pleading) "Cindy, please stop! This isn't right!" Amy whispered words of encouragement to John while shifting her body forward. The movement caused the ropes to dig deeper into her flesh, but she forced herself closer to John. [Amy] (hoarse whisper) "You can do it, John. I believe in you."

Good Ending

John moved swiftly, knocking Cindy's knife away with the lamp. At that moment, Noah pulled Cindy back with all his strength. John wasted no time, rushing to Amy and freeing her. Cindy collapsed into Noah's arms, sobbing. John and Amy embraced tightly. By the time the police arrived, it was all over.

Months later, Cindy was admitted to a psychiatric hospital for treatment, and Noah remained by her side, dedicated to helping her recover. John and Amy, having grown closer through the ordeal, promised to marry. They frequently visited Noah to check on Cindy's condition. Over time, all four of them slowly

healed from the trauma and moved forward. Amy, inspired by her experience, founded an organization to support crime victims, with John actively assisting her. Though the event had changed their lives forever, it also strengthened their bond and showed them the power of love and resilience. The End.